Tender No: BMRCL/Phase-3/JPN-MRS/Demolition/DLN-P1/2026/157

BANGALORE METRO RAIL PROJECT, PHASE-3

Name of the Work: "Partial and Full Demolition/Dismantling of Buildings/Structures identified in Bangalore Metro Rail Project Phase-3."

Contract Package-P1: "Partial and Full Demolition/Dismantling of Buildings/Structures identified from JP Nagar 4th Phase Station to Mysore Road Station (from Ch: (-) 0.330 Km to Ch: 10.550 Km) in Corridor-1of Bangalore Metro Rail Project Phase-3."

TECHNICAL DOCUMENT

VOLUME -2

SEPTEMBER-2026

SECTION – H - EMPLOYER'S REQUIREMENT AND TECHNICAL SPECIFICATONS

Table of Contents

PART - I	**GENERAL**	**2**
PART - 2	**THE KEY ENGINEERING CONTROLS AND WORK PRACTICES**	**6**
PART- 3	**SCOPE OF WORK**	**12**
Annexure-1 to Employer's Requirement	**MILESTONES (KEY DATES) AND LIQUIDATED DAMAGES**	**17**
Annexure-2 to Employer's Requirement	**MINIMUM QUALIFICATION AND EXPERIENCE REQUIRED FOR KEY PERSONNEL TO BE DEPLOYED FOR THE WORK**	**18**
Annexure-3 to Employer's Requirement	**MINIMUM PLANTS & EQUIPMENTS FOR THE WORK**	**19**
PART -4	**TECHNICAL SPECIFICATIONS**	
	PART A : SPECIFICATIONS FOR DEMOLITIONS/ DISMANTLING WORKS	**20**
	PART B: SPECIFICATIONS FOR REHABILITATION AND RESTORATION WORKS	**30**

PART 1 - GENERAL

1. The Employer's Responsibility:

1.1 The Employer shall grant the Contractor right of access to, and possession of, the Site progressively for the completion of Works. Such right and possession may not be exclusive to the Contractor. The Contractor will draw/modify the schedule for completion of Works according to progressive possession/right of such sites.

1.2 If the Contractor suffers delay from failure on the part of the Employer to grant right of access to, or possession of the Site, the Contractor shall give notice to the Engineer in a period of 28 days of such occurrence. After receipt of such notice the Engineer shall proceed to determine any extension of time to which the Contractor is entitled and shall notify the Contractor accordingly.
For any such delay in handing over of site, Contractors will be entitled to only reasonable extension of time and no monetary claims whatsoever shall be paid or entertained on this account.

1.3 The Employer may (Where he is in a position to do so) provide reasonable assistance to contractor in getting permits, license on approvals during contract.

1.4 The successful tenderer will be issued with "Demolition/ Dismantling" scheme drawings on award of contract.

2. The Contractor's Responsibility:

2.1 The Works as completed by the Contractor shall be wholly in accordance with the Contract and fit for the purposes for which they are intended, as defined in the Contract.

2.2 The Contractor shall provide all superintendence, labour, machinery etc.

2.3 The Contractor shall take full responsibility for the adequacy, stability and safety of all methods of dismantling/demolition, irrespective of any approval or consent by the Engineer/consultant

2.4 The Contractor acknowledges responsibility for ascertaining and securing at his own cost:

(a) Conditions bearing upon the proper transportation, disposal, handling and storage of materials (including but not limited to hazardous toxic substances and excavated materials);
(b) Availability of electricity, water and gas;
(c) Availability of skilled manpower;
(d) The protection of the environment and adjacent structures.
(e) The location of and the authorization required for and the means of diversion of any services and facilities required for the purposes of the Works.

2.5 The Contractor shall not impede and shall afford all necessary facilities, access and/or services to the Employer, Engineer, Designated Contractors, utility undertakings, other relevant authorities and other contractors (whether employed by the Employer or not) who are carrying out duties, on or in the vicinity of the Site not included in the Contract but forming part of the Project.

3. The Consultant's/Engineer's Responsibility:

The Employer may nominate a Consultant/Engineer for executing this work. The responsibilities of the Consultants/Engineer are given below in brief:

(i) Suggest the type and capacity of equipment required for safe demolition process. Prepare structure wise report on sequence and methodology of demolition including remedial measures required to be undertaken prior to demolition, during demolition and post demolition.

(ii) Suggest legal and safe disposal of demolished/dismantled items in which land loser has no interest, to the designated places identified by the contractor, except debris.

The debris shall be disposed of at BBMP/GBA identified C&D waste management facility.

(iii) Suggest the restoration works to be done, from the point of safety and functional use.

(iv) Prepare design and drawings for temporary structures including safety barricading to be used during the demotion of work by the demolition contractor.

(v) Supervise the demolition and restoration works implemented through the demolition contractor and ensure that the demolition contractor strictly follows the demolition/dismantling methodology, with all safety standards.

vi) Certification of bills submitted by contractor for the work done.

4. General Principles of Demolitions/Dismantling of the Buildings/Structures.

The Contractor shall adopt appropriate methodology and equipment for safe demolition/dismantling of the buildings/ Structures. Certain precautions not limited to the following are required to be taken during demolition work:

(a) Prior to start demolition operations, the contractor shall conduct an engineering survey and survey report shall be discussed with consultant/Engineer appointed by the employer for consensus and approval of the consultant /Engineer/BMRCL. In engineering survey a competent person preferably structural engineer should inspect the site to determine the condition of the framing, floors and walls, and possibility of unplanned collapse of any portion of the structure. Any adjacent structure shall also be similarly checked before commencement of demolition.

(b) Within a structure to be demolished which has been damaged by fire, flood, explosion, or other cause, the walls or floor shall be shored or braced.

(c) The contractor should ensure that. all electric, gas, water, steam, sewer and other service lines are shut off, capped, or otherwise controlled (at his own cost), outside the building line before starting the demolition work. If any notification is to be issued by BMRCL, BMRCL will issue the notification to the concerned.

(d) If it is necessary to maintain (temporarily) electric power, water or other utilities during demolition, such lines shall be temporarily relocated, as necessary and protected before commencement of demolition work. The contractor shall make his own arrangements for power requirement for his equipment/machinery used for demolition.

(e) It shall also be determined if any type of hazardous chemicals, gases, explosives, flammable materials, or similarly dangerous substances have been used in any pipes, tanks, or other equipment on the property. When the presence of any such substances is apparent or suspected, testing and removal shall be performed and the hazard eliminated before demolition is started.

(f) Where a hazard exists from fragmentation of glass or any such kind of materials hazards shall be removed.

(g) Where a hazard exists to contractor’s employees and the residents falling through wall openings, the opening shall be protected to a height of approximately 105 cm.

(h) When debris is dropped through holes in the floor without the use of chutes, the area onto which the material is dropped shall be completely enclosed with barricades not less than 100cm high and not less than 180cm back from the projected edge of the opening above. **Signs, warning of the hazard** of falling materials, shall be posted at each level. Removal shall not be permitted in this lower area until debris handling ceases above.

(i) All floor openings, not used as material drops, shall be covered over with material substantial enough to support the weight of any load which may be imposed. Such material shall be properly secured to prevent its accidental movement.

(j) Except for the cutting of holes in floors for chutes, holes through which to drop materials, preparation of storage space, and similar necessary preparatory work, the demolition of exterior walls and floor construction shall begin at the top of the structure and proceed downward. Each story of exterior wall and floor construction shall be removed and dropped into the storage space before commencing the removal of exterior walls and floors in the story next below.

(k) Employee entrances to multistory structures being demolished shall be completely protected by sidewalk sheds or canopies, or both, providing protection from the face of the building for a minimum of 2.4m. All such canopies shall be at least 0.6m wider than the building entrances or openings (0.3m wider on each side thereof), and shall be capable of sustaining a load of 750kg per square meter.

(l) The contractor shall take sufficient care to erect **safety sign boards** and sufficient to manage traffic flow during demolition should be ensured. If required, local traffic police to be informed prior to commencement of demolition.

(m) **Safety procedure:** The demolition contractor shall:

(i) Comply with all applicable safety regulations.
(ii) Take care for the safety of all persons entitled to be on site. Provide Personal Protective Equipment to all workers involved. PPEs shall include Helmets, Safety Shoes, Safety belts, Goggles, Hand Gloves, Safety Jackets and other equipment required for safety of work force.
(iii) Use reasonable efforts to keep the site and works clear of unnecessary obstructions so as to avoid dangers to persons.
(iv) Provide fencing, lighting, guarding and watching of the works until completion of the demolition and safety certification (by the structural engineer) of remaining portion of the building where partial demolition is carried out.
(v) Provide any temporary works including road ways, foot ways, guards and fences which may be necessary because of the demolition of the building, for the use and protection of public and of owners and occupiers of adjacent buildings.
(vi) The structure to be demolished/dismantled must have barricades, which shall be of plain MS Sheet of 16 gauge fixed with steel frame, painting, with synthetic enamel paint of approved color, quality and brand, writing letter and logo of metro including maintenance of the same, duly cleaning the same frequently, arrangement for blinker light as per requirement. During demolition barricades has to be kept continuously. Removal of barricade will not be permitted till completion of demolition, dismantling of properties.

(n) **Protection of the Environment:** The contractors shall take all reasonable steps to protect the environment (both on and off the site) and to limit the damage and nuisance to the people and property resulting from pollution, noise and other result of the operations. The contractors shall ensure that emissions, surface discharges and effluent from the contractor's activities shall not exceed the values prescribed by the applicable laws. The dust and noise level during demolition to be maintained within acceptable norms under the provisions of BMRCL safety manual/applicable codal standards including regular watering for dust suppression.

(o) **Security of the site:** The contractor shall be responsible for keeping unauthorized person off the site and authorized person shall be limited to the contractor's personnel and BMRCL personnel and to any other personnel notified to the contractor by the BMRCL.

(p) **Contractor's operations on site:** The contractor's shall confine his operations to the site and to any additional areas which may be obtained by the contractor, agreed by the BMRCL as working area. The contractor shall take all necessary precautions to keep contractor's equipment and contractor's personnel within the site.

(q) **Labour laws:** The Contractor shall comply with all the relevant labor laws applicable to the contractor's personnel, including laws relating to their employment, health, safety, welfare, immigration and emigration and shall allow them all their legal rights. The Contractor shall require his employees to obey all applicable laws, including those concerning safety at work.

(r) **Health and Safety:** The Contractor shall at all times take all reasonable precautions to maintain the health and safety of the Contractor's Personnel. In collaboration with local health authorities, the Contractor shall ensure that medical staff, first aid facilities, sick bay and ambulance service are available at all times at the site and any accommodation for Contractor's and Employer's Personnel and that suitable arrangements are made for all necessary Welfare and hygiene requirements and for the prevention of epidemics.

The Contractor shall appoint an accident prevention officer at the site, responsible for maintaining safety and protection against accidents. This person shall be qualified for this responsibility and shall have the authority to issue instructions and take protective measures to prevent accidents. Throughout the execution of the works, the Contractor shall provide whatever is required by this person to exercise this responsibility and authority.

The Contractor shall send, to the BMRCL details of any accident as soon as practicable after its occurrence. The Contractor shall maintain records and make reports concerning health, safety and welfare of persons and damage to property as the Engineer may reasonably require.

PART 2- KEY ENGINEERING CONTROLS AND WORK PRACTICES

The following key engineering controls and work practices are briefed for the guidance of the contractor for achieving safety standards during planning operation and closing of the task of Demolition /Dismantling buildings and structures.

1. General Recommendations:

In addition to general principals to be follow as explained in Para's above the Personal Protective Equipment (PPE) the contractor is encouraged the following additional PPE that may be needed for a specific hazard is noted below.

General PPE includes:

- Steel Helmets/Safety helmets for overhead impact or electrical hazards.
- Safety Belts and safety jackets.
- Eye protection with side shields
- Gloves chosen for job hazards expected (e.g., heavy-duty leather work gloves for handling debris with sharp edges and/or chemical protective gloves appropriate for chemicals potentially contacted)
- Approved protective footwear
- Respiratory protection as necessary—filtering face pieces may be used for nuisance dusts (e.g., dried mud, dirt and silt)

2. STRUCTURAL INSTABILITIES

Key Engineering Controls and Work Practices

- Conduct an engineering survey of the framing, floors, and walls of the structure and any adjacent structures that might be affected; maintain written survey results
- Shore/brace walls and floors as needed to maintain safe work areas during demolition activities
- Have a competent person continually inspect work area to detect hazards resulting from weakened or deteriorated floors, or walls, or loosened material. A competent person is able to recognize existing and predictable hazardous conditions and has the authority to take prompt corrective measures to eliminate the hazardous condition
- Do not allow falling debris/stored materials or heavy equipment to exceed the safe carrying capacity of the floor
- Conduct demolition work from the topmost floors to down. Do not cut / remove load-supporting members until upper or supported loads have been removed

3. UNSECURED HAZARDS IN THE WORK AREA

Key Engineering Controls and Work Practices

- Remove or secure objects (glass, structural members) that may fall while employees work under them.
- Use debris netting, sidewalk sheds, canopies, or catch platforms to reduce hazards from falling objects
- Verify the location of all other utility lines; ensure lines have been shut-off, capped, or otherwise controlled outside the building before beginning work. Notify utility companies before controlling their utility lines
- Assess the presence, contents, and condition of tanks and equipment that might contain hazardous chemicals, gases, or flammable materials. If the condition of tanks or equipment is suspect, avoid disturbing them until after the assessment is done and a plan of action determined

- Purge lines, tanks, and equipment containing hazardous chemicals, gases, or flammable materials. Use air monitoring equipment such as a combustible gas indicator, toxic gas monitor, or oxygen monitor to determine if any hazardous conditions remain. When necessary, provide additional controls to protect response and recovery workers (e.g., forced ventilation, respiratory protection)
- Provide and use safe walkways to reach any point without having to walk on exposed beams; walkways should be at least 45 cm wide and formed by using 5 cm thick wood, with stringers installed as needed for support

4. DROPPING CONSTRUCTION DEBRIS

Key Engineering Controls and Work Practices

- Enclose material chutes at greater than a 45 degree angle; chute openings should be 120 cm or less with gates that are closed when the chute is not in use; provide 105 cm high guardrails where debris is dumped manually; where debris is dumped using mechanical equipment or wheel barrows, provide 10 cm wide by 15 cm high toe boards/bumpers
- Barricade and mark all debris-dropping areas; ensure debris is not removed until debris-handling ceases above

5. IMPROPER LADDER OR SCAFFOLD USE

Key Engineering Controls and Work Practices

- Inspect ladders for cracked, broken, or defective parts before use
- Do not exceed the load rating of ladders or scaffolds remember that load ratings include people, tools, and equipment
- Set up ladders and scaffolds on stable surfaces
- Set extension or straight ladders at a 75 degree angle from the ground (7 cm back for every foot of rise) and provide 90cm above an upper landing surface to ease climbing onto/descending from height
- Use non-conductive ladders (e.g., fiberglass) and exercise extreme caution when working near power lines
- Secure ladders that can be displaced by work activities; consider barricades at the base to keep traffic away
- Have a competent person inspect scaffolds before use
- Ensure that the scaffold is plumb, and braced and guyed to prevent tipping, swaying, and displacement
- Ensure that the scaffold is built on base plates and mud sills, or other firm foundations. Footings should be able to support the scaffold without settling or moving. Do not use unstable objects to support scaffolds
- Fully plank each scaffold on all working levels. For wood planking, use wood graded for the intended load
- Provide guardrails or fall protection systems on platforms 3.3m or higher

Additional Personal Protective Equipment

- Fall arrest systems on platforms without guardrails 3.3 m or higher

6. GENERAL HEAVY EQUIPMENT OPERATIONS

Additional Personal Protective Equipment

- Hearing protection for Noise hazard

- When working from an aerial lift, use a body harness that is properly attached (or body belt for tethering or restraint use only) for fall protection

7. FALLS FROM HEIGHTS OR THROUGH OPENINGS

Key Engineering Controls and Work Practices

- Limit access/set up controlled access zones
- Use fall protection systems: guardrails, safety nets, or fall arrest systems
- Cover or guard holes and openings as soon as they are created. Covers must support two times the weight (body, equipment, materials) that may be imposed

Additional Personal Protective Equipment

- Personal fall arrest system including harnesses, lanyards, lifelines, connectors, anchorages, and anchor points (as needed)

8. CONTACT WITH LIVE ELECTRICAL EQUIPMENT AND OTHER UTILITIES

Key Engineering Controls and Work Practices

- Assume that electrical lines are energized until proven otherwise. Lines and other conductors may become reenergized without warning as utilities are evaluated and restored after a disaster
- Inspect the work area for downed conductors and do not go near, drive over, or otherwise come in contact with them
- Downed electrical conductors can energize other objects, including fences, water pipes, bushes, trees, and telephone/CATV/fiber optic cables
- Unless de-energized and visibly grounded, maintain proper distance from overhead electrical power lines (at least 3.3m) and/or provide insulating barriers
- Do not approach any gas leaks; if a gas leak is detected, secure spark-producing devices (e.g., engines, tools, electronic and communications equipment) and evacuate the area until the leak is secured

9. POWER AND HAND TOOL USE

Key Engineering Controls and Work Practices

- Use ground-fault circuit interrupters (GFCIs) or double insulated power tools, or implement an assured equipment grounding program
- Inspect power tool condition (including any cords) and verify operation of safety features before use
- Do not use equipment that is defective, such as equipment with inoperable safety switches, missing guards, frayed/cut cords, etc.
- Ground power tools properly
- Avoid standing in wet areas when using portable power tools

Additional Personal Protective Equipment

- Hearing protection for Noise hazard
- Hand protection for cut- and abrasion-control and vibration dampening
- Eye protection appropriate to the impact hazard

10. GENERATOR USE

Key Engineering Controls and Work Practices

- Never attach a generator directly to the electrical system of a structure unless a qualified electrician has installed a transfer switch for the generator. If the structure's electrical system is not isolated, it may energize the utility's wiring system for great distances and create a risk of electrocution for utility workers and others in the area
- Always plug electrical equipment directly into the generator using the manufacturer's supplied cords or grounded (3-pronged) extension cords that are rated for the total anticipated load
- Do not overload a generator; it can overheat and create a fire hazard
- Ground and bond generators according to the manufacturer's recommendations; ensure that any manufacturer-required connections are secure before using the generator
- Keep the generator dry; protect with a canopy if needed; do not use it in wet or rainy conditions
- Carbon monoxide (CO) is a poisonous, colorless, and odorless gas that is produced by the incomplete burning of the generator's fuel. CO is harmful when breathed because it displaces oxygen in the blood and deprives the heart, brain, and other vital organs of oxygen
- Never use a generator indoors or in enclosed spaces such as garages and basements; opening windows and doors may not prevent CO from building up in those spaces. Do not use a generator outdoors near doors, windows, and vents that could allow CO to enter
- Ensure that a generator has 1 to 1.2 m of clear space on all sides and above it to ensure adequate ventilation and cooling. Before refueling, shut down the generator and allow it to cool

Additional Personal Protective Equipment

Hearing protection for Noise hazard.

11. SILICA, MOLD, NUISANCE DUST, DRIED MUD, OR SILT

Key Engineering Controls and Work Practices

- Stay upwind of or away from dust-generating activities, and in particular those involving crystalline silica-containing materials like concrete, brick, tile, drywall, mortar, sand, or stone. When inhaled, the fine crystalline silica particles contained in the dust can become lodged deep in the lung, which can lead to silicosis and other respiratory illnesses
- Use water spray or mist to suppress dust generation, especially during operations that may create a lot of dust, such as cutting or sawing silica-containing materials, jack hammering, impact drilling, using heavy equipment, and demolishing structures
- Avoid using compressed air for cleaning surfaces
- Limit contact or disturbance of surfaces containing substantial visible mold growth

Additional Personal Protective Equipment

- At a minimum, use respirators with filters for work with crystalline silica-containing materials (e.g., concrete, brick, tile, mortar). The use of filters may provide additional protection. Higher levels of respiratory protection may be needed for some operations (e.g., cutting concrete, sandblasting, mixing concrete)
- Respirators may be used for nuisance dusts (e.g., dried mud, dirt, or silt)

12. WELDING, CUTTING, AND BURNING

Key Engineering Controls and Work Practices

- Remove flammable and combustible materials from the area
- Do not perform "hot work" such as welding, cutting, or burning in areas where flammable, combustible, corrosive, or toxic substances are being used, stored, or may otherwise be present
- Maintain a fire watch during all hot work until material has cooled
- Ensure fire extinguishers and extinguishing agents are available in the immediate area
- Provide natural, exhaust, or forced ventilation to control exposure to the metal fumes and other contaminants being generated (e.g., generator exhaust)
- Ensure that pipes and other vessels are purged of hazardous materials
- Identify building materials that will be welded, cut, or burned and that may contain lead, such as painted surfaces and pipes.

Additional Personal Protective Equipment

- Gloves and protective clothing for the activity being performed
- At a minimum, filtered lenses and face-protection as appropriate for the activity being performed
- Respiratory protection based on anticipated exposure to metal fumes, including lead

13. NOISE

Key Engineering Controls and Work Practices

- Place generators, compressors, and other noisy equipment at a distance or behind a barrier when possible

Additional Personal Protective Equipment

- Hearing protection when working around potential noise sources and when noise levels exceed 90 dba. A useful "rule of thumb"—if you cannot hold a conversation in a normal speaking voice with a person who is standing at arm’s length (approximately 0.9m), the noise level may exceed 90 dba.

14. CONFINED SPACES

Key Engineering Controls and Work Practices

- Confined spaces have limited means of entry or exit, are large enough to bodily enter, and may contain physical (e.g., mechanical, electrical, hydraulic, pneumatic energy; engulfment hazards; inwardly converging surfaces) or atmospheric hazards (e.g., atmospheres that are oxygen-deficient or oxygen-enriched, contain or may contain flammable gas, vapor or mist, airborne combustible dust, toxic substances, or any other atmosphere that is immediately dangerous to life or health). Examples include storage tanks, process vessels, bins, boilers, vaults, ventilation or exhaust ducts, sewers, tunnels, pipelines, and pits more than 1.2m in depth
- Some events might introduce hazards or potential hazards into confined spaces. For example, a space might have a potential to contain a hazardous atmosphere due to the presence of decomposing organic matter, to the use of hazardous chemicals in the space, or to the performance of operations in the space, such as welding, cutting,

or burning, that may create a hazardous atmosphere. Additional precautions must be taken to make the space safe for entry

- Evaluate the need for entry (i.e., placing any body part into the space); prevent unauthorized entries

15. BARRICADING THE PROPERTIES DURING DEMOLITION/DISMANTLING

The structure to be demolished/dismantled must have barricades, which shall be of plain MS Sheet of 16 gauge fixed with steel frame, painting, with synthetic enamel paint of approved color, quality and brand, writing letter and logo of metro including maintenance of the same, duly cleaning the same frequently, arrangement for blinker light as per requirement. During demolition barricades has to be kept continuously. Removal of barricade will not be permitted till completion of demolition, dismantling of properties

.

PART-3: SCOPE OF WORK

1. General Requirements

(a) Contract Package – P1: "Partial and Full Demolition/Dismantling of Buildings/Structures identified from JP Nagar 4th Phase Station to Mysore Road Station (from Ch: (-) 0.330 Km to Ch: 10.550 Km) in Corridor-1of Bangalore Metro Rail Project Phase-3."

To commence the civil construction works and float the civil tenders, demolition/dismantling of the buildings and structures identified for the viaduct, double-decker structure, stations, loops, and ramps shall be undertaken.

This package includes the demolition/dismantling of various types of commercial and residential buildings ranging from Basements, Ground Floor (G+0) to Ground +4 (G+4), as well as sheds. The scope of work includes demolition of the identified structures, salvage of specified items and materials, and removal and disposal of the resulting debris, walls with all leads and lifts, etc. The debris shall be disposed of at BBMP/GBA identified C&D waste management facility.

The identified buildings and structures have been classified into the following categories:

1. Partial demolition/dismantling of all types of structures of varying heights, consisting of basements, ground floor and/or one or more upper floors (RCC framed structures, load-bearing structures with RCC roofs, and other structures such as AC roof structures, etc.), by mechanical and manual methods, comprising cement concrete, stone masonry, brickwork, steelwork, woodwork, etc., with a total quantity as mentioned in the BoQ.

2. Full demolition/dismantling of all types of structures of varying heights, consisting of basement level, ground floor and/or one or more upper floors (RCC framed structures, load-bearing structures with RCC roofs, and other structures such as AC roof structures, etc.), by mechanical and manual methods, with a total quantity as mentioned in the BoQ.

3. Demolition/dismantling of compound walls of all shapes and heights by mechanical and manual methods, comprising cement concrete, stone masonry, brickwork, steelwork, fencing, stone slab walls, chain-link fencing, etc., with a total quantity as mentioned in the BoQ.

4. Demolition/dismantling of steel structures such as bus stops, median grills, compound wall grills, signboards, hoardings, etc., with a total quantity as mentioned in the BoQ. Dismantled/released material to be handed over to the concerned authority at the locations specified by owning authority and removal of resulting debris with all lead and lifts and as directed by the Engineer.

(b) Demolition and dismantling works.

The work includes demolition, salvage of identified items and materials, and removal of resulting debris. Debris shall be removed from the premises daily, unless otherwise directed, to avoid accumulation at the demolition premises. The demolition shall be carried out after taking necessary strengthening methods for the buildings proposed for demolition and also for neighboring buildings. The contractor shall develop and take necessary strengthening methods for the balance portion of the structure not being dismantled. In the interest of occupational safety and health, the work shall be performed

in accordance with relevant IS codes on demolition and other applicable Sections. In the interest of conservation, salvage shall be pursued to the maximum extent possible. The salvaged items and materials shall be disposed of as specified in which the 'owner losing structure totally/partially' and the 'contractor' has no interest. The work also includes disconnection of power supply with the permission of BESCOM, rewiring and shifting of energy meter along with accessories to the suitable place as per BESCOM's direction for partially demolished building. For full demolition of buildings, disconnection of power supply, releasing and handing over the electrical energy meter along with accessories as per directions of concerned BESCOM office through govt. licensed electrician.

(c) Restoration works for safety reasons.

The BMRCL has appointed consultant/Engineer who has conducted Building Condition Study (BCS) and suggest sequence and methodology of demolition including remedial measures required to be undertaken prior to demolition and / or during demolition and / or post demolition.

It is anticipated that, certain structures will be required to be constructed for safety and functional reasons as a result of Building **Condition** Study. The structures anticipated are front walls, doors, windows, beams and columns etc.

Works done as permanent remedial measures (for example –excavation, backfilling, PCC, RCC, brick/RR/CC block masonry, fixing of doors, windows, plastering, painting of face of wall remained after demolition, water supply, sanitation, electrical works including restoration of power supply to the balance building portion through govt. licensed electrician, etc.) shall be paid as per accepted price for the BoQ schedule for KPWD-Common SR 2023-24. The contractor shall carry out these works after obtaining instructions from Consultant / Engineer/BMRCL to that effect. The contractor has to execute these items in accordance with KPWD-Common SR 2023-24 and rate in the accepted priced bill of quantities. Measurements will be recorded for only those buildings completed under the supervision of consultant/Engineer appointed for Building Condition Study.

2. Work Plan. The procedures shall provide for safe conduct of the work, including methods to provide necessary supports, lateral bracing and shoring when required, careful removal and disposition of materials to be salvaged, protection of property, which is to remain undisturbed, coordination with other work in progress, and timely disconnection of utility services. The procedures shall include a detailed description of methods and equipment to be used for each operation, and the sequence of operations.

The contractor in addition to the proposed methodology should include the following:

- Initial Engineering survey of the building to be done.
- Developing strengthening methods of the balance portion of the structure not being dismantled and neighboring buildings.
- The actual method of dismantling proposed to be adopted.

3. Dust Control. The dust resulting from demolition shall be controlled to prevent the spread of dust to occupied portions of the construction site and to avoid creation of a nuisance in the surrounding area. Use of water will not be permitted when it will result in, or create, hazardous or objectionable conditions such as flooding and pollution. The contractor shall take all measures to bring down dust level and cause least pollution in air quality and take necessary measures to keep the noise pollution as low as possible.

4. Protection of Personnel. During the demolition work the contractor shall continuously evaluate the condition of the structure being demolished and take immediate action to protect all people working in and around the demolition site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent

collapse or failure while workmen remove debris or perform other work in the immediate area. The contractor shall use wire nets etc., liberally so that falling debris etc., will not cause any injury to people in the premises. The methods to be adopted for dismantling/demolition of the structures should be apprised in advance to BMRCL.

5. **Protection of Structures.** Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, shall remain standing without additional bracing, shoring, of lateral support until demolished, unless directed otherwise by the contracting officer. The contractor shall ensure that no elements determined to be unstable are left unsupported and shall be responsible for placing and securing bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

6. **Protection of Existing Property.** Before beginning any demolition work, the contractor shall survey the site and examine the drawings and specifications to determine the extent of the work. The contractor shall take necessary precautions to avoid damage to existing items to remain in place, to be reused or to remain the property any damaged items shall be repaired or replaced as approved by the Engineer. The contractor shall coordinate the work of this section with all other work and shall construct and maintain that structural elements are not overloaded and shall be responsible for increasing structural supports or adding new supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

7. **Protection from the weather.** Salvageable materials and equipment shall be protected from the weather at all times.

8. **Protection of Trees.** Trees within the project sites which might be damaged during demolition and which are indicated to be left in place shall be protected by a high fence. The fence shall be securely erected a minimum of 2 m from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Any tree designated to remain that is damaged during the work under this contract shall be replaced in kind or as approved by the contracting officer.

9. **Environmental Protection.** The work shall comply with the requirements of relevant environment protection clauses.

10. **Burning.** Burning at the project site for disposal of refuse and debris is not permitted.

11. **Use of explosives.** Use of explosives will not be permitted

12. **Utilities.** Existing utilities shall be removed as indicated. When utility lines are encountered that is not indicated on the drawings, the contractor shall be notified prior to further work in that area, and shall follow instruction.

13. **Filling.** Holes, open basements and other hazardous openings shall be filled in accordance with relevant IS codes.

14. **Disposal of material.** The Contractor shall dispose the demolished materials except Government Salvage and historical items, which shall be duly informed to the Engineer and shall be handed over to the appropriate authorities as per the procedure in vogue. The rates quoted by the contractor includes loading, unloading and stacking with all lead and lifts to dispose the demolished materials.

15. **Salvageable items and material.** As per Compensation and Resettlement guidelines of BMRCL, the owner losing land and structure has the right to salvage materials totally. In the case of items required by the 'owner losing structure totally/partially', he shall remove the same from the site before the contractor demobilizes from the site. The salvage materials in which the 'owner losing structure totally/partially' is not interested shall be owned by the Contractor and the value of the same shall be credited to BMRCL to the maximum extent possible.

16. **Material Salvaged for the contractor.** The contractor shall not sell the salvaged materials at site.

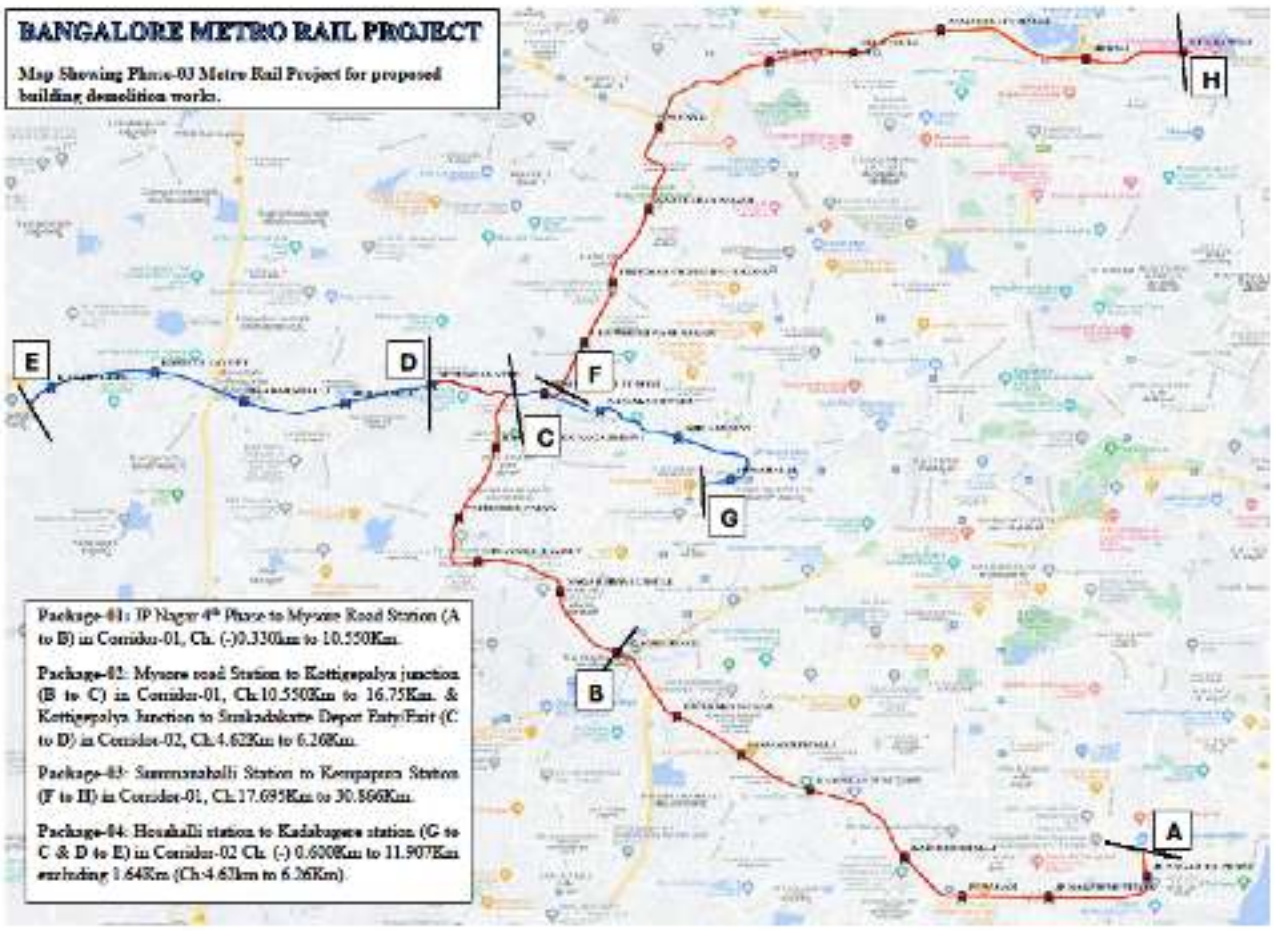
17. **Item salvaged for the Government.** In the case of salvaged materials of the Government, the contractor shall remove the salvaged materials in a manner that there is no damage while packing or carting and protect the materials from damage during storage or transportation. In case of any damage during storage or transportation, the same shall be repaired or replaced to match the existing items. Dismantled/released materials to be handed over to the concerned authorities at the locations specified by the owning authority.

18. **Clean up.** Debris and rubbish shall be removed from basement, ground and any other stacking place. Debris shall be removed and transported in a manner that prevents spillage on streets or adjacent areas and no spillage is allowed during transportation.

19. **Handing over of work front.** List of buildings to be demolished including area of demolition etc. will be made available to the contractor by BMRCL progressively.

20. **Barricading the properties during demolition / dismantling.** The structure to be demolished/dismantled must have barricades, which shall be of plain MS Sheet of 16 gauge fixed with steel frame, painting, with synthetic enamel paint of approved color, quality and brand, writing letter and logo of metro including maintenance of the same, duly cleaning the same frequently, arrangement for blinker light as per requirement. During demolition barricades has to be kept continuously. Removal of barricade will not be permitted till completion of demolition, dismantling of properties.

21. Demolition of buildings/structures is to be done by ensuring minimal vibration, without affecting the stability/serviceability of the remaining structure after demolition. Dismantling of structure is permitted only by using diamond disc cutter/floor saw machine at all levels. However, the decision of Engineer in charge is final and binding on the contractor. The structural integrity of the remaining structure will be analyzed by structural experts engaged by Employer/Engineer by visual observation and necessary Non-Destructive Tests. In case any defect is noticed on account of demolition / dismantling activity and/or any claim is raised by the owner of the building/structure, the damage will be repaired and claim, if any, shall be settled by the contractor, failing which the defects will be rectified at the risk and cost of the contractor as per the directions of Engineer/Employer and the same will be recovered from the contractor. Necessary scaffolding to be provided to facilitate the dismantling of buildings/structures as mentioned in the line-of-cut proposed by structural experts or as directed by the Engineer In-charge. The scaffolding shall have to be erected with proper anchors to ensure safety & stability during demolition / dismantling.


BANGALORE METRO RAIL PROJECT
Map Showing Phase-03 Metro Rail Project for proposed building demolition works.
Package-01: IP Nagar 4th Phase to Mysore Road Station (A to B) in Corridor-01, Ch. (-)0.330km to 10.550Km.
Package-02: Mysore road Station to Kottigepalya junction (B to C) in Corridor-01, Ch.10.550Km to 16.75Km. & Kottigepalya Junction to Srirakshakatte Depot Entry/Exit (C to D) in Corridor-02, Ch.4.62Km to 6.26Km.
Package-03: Sampangiramanagar Station to Kempapura Station (F to H) in Corridor-01, Ch.17.695Km to 30.866Km.
Package-04: Hosahalli station to Kadabagere station (G to C & D to E) in Corridor-02 Ch. (-) 0.600Km to 11.907Km excluding 1.64Km (Ch:4.62km to 6.26Km).
A
B
C
D
E
F
G
H

ANNEXURE -1 to EMPLOYER'S REQUIREMENT

MILESTONES (KEY DATES) AND LIQUIDATED DAMAGES

i. Milestones:

Sl. No	**Milestone**	**Period from the start date**	**No. of buildings / structures to be demolished/dismantled**	**Liquidated damages**
1.	Milestone – 1	Six months from the date of signing the contract agreement.	50% of total area **handed over**.	Rs. 5,000/- per day for delay in achieving Milestone-1
2.	Milestone – 2	Twelve months from the date of signing the contract agreement.	100% of total area **handed over**.	Rs. 10,000/- per day for delay in achieving Milestone-2

ii. The upper limit for Liquidated Damages is 5% of Contract Price.

ANNEXURE -2 to EMPLOYER'S REQUIREMENT

MINIMUM QUALIFICATION AND EXPERIENCE REQUIRED FOR KEY PERSONNEL TO BE DEPLOYED FOR THE WORK

(To be submitted by Tenderer)

Sl. No	Key Personnel	Qualifications & Total Experience (Post Qualification)	Experience in Similar Nature works (Minimum requirement) (Post Qualification)	Minimum Number of Personnel Required	Name of the key personnel proposed	Qualification	Total number of years of experience (Post Qualification)	Number of Years of experience in the proposed position / equivalent in Similar Nature works (Post Qualification)
1	Project Manager	BE /B. Tech in Civil Engineering with 10 years of experience (Post Qualification)	3 years as Project Manager or equivalent in Similar Nature Works	1				
2	Chief Safety and Health Manager	Graduate in Engineering / MSc in Safety from a recognized university with 8 years of total experience (Post Qualification) .	5 years as Chief Safety and Health Manager or equivalent in Infrastructure projects	1				

ANNEXURE -3 to EMPLOYER'S REQUIREMENT

(To be submitted by Tenderer)

MINIMUM PLANTS & EQUIPMENTS FOR THE WORK

(TO BE DEPLOYED AS AND WHEN REQUIRED)

Tenderers should have the following plant and machinery either owned or leased or hired by him.

Sl. No	Type of equipment required for the work	Proposed to be deployed
1	Diamond disc Cutter/floor saw machine with suitable capacity	3 Nos
2	Wall Sawing Machine with suitable capacity	3 Nos
3	Electric Drill Motors/ Hammer drilling machine	4 Nos
4	Floor Saw Machine with suitable capacity	4 Nos
5	Chipping Machine	12 Nos
6	Crusher/Combi Crusher	2 Nos
7	Hydraulic Mobile crane	2 Nos
8	High/Long Reach Demolition equipment as per the requirement	1 Nos
9	Hydraulic rock breaker with loader	2 Nos
10	DG set 100kV or higher capacity	2 Nos

NOTE:

1. Plant and Equipment indicated above is minimum to be deployed at appropriate stage of the work. However, depending on the requirement to complete the work in the stipulated completion period, the tenderer should deploy additional machinery as circumstances warrant at no extra cost.
2. Plant and equipment to be mobilized for the work shall be in good serviceable condition.

PART – 4: TECHNICAL SPECIFICATIONS

PART – A – SPECIFICATIONS FOR DEMOLITION / DISMANTLING WORKS

1. Dismantling. The term "dismantling" implies carefully taking up or down and removing the building materials without damaging them. The articles dismantled shall be lowered to the ground and not thrown. Dismantling work shall cover complete removal of the existing structure or part of a work including all relevant items as indicated or as directed, clearing the site, sorting out useful materials and stacking them as described, and disposing of the unserviceable materials.

2. Demolition. The term "demolition" implies breaking up the components of the structure building and then taking the components up or down. This shall consists of demolishing whole or part of work including all relevant items as indicated or as directed, clearing the site, sorting our useful materials and stacking them as directed and disposing of the unserviceable materials and rubbish as directed. The removal of overlaying or adjacent materials, if required for demolition of the structure shall be separately indicated. Unless otherwise specified, the building/structure shall be dismantled/ demolished up to ground level or up to basement level.

3. Serviceable and unserviceable materials

3.1 **Inventory.** Before dismantling/demolition operations are undertaken by the contractor, inventory of all materials, fittings and fixtures (except hidden materials) which are considered useful shall be made and signed by the Employers representative and the contractor, wherever the operations are entrusted to a contractor.

3.2 **Serviceable materials.** Any material which is in the opinion of the Employer's representative could be reused or otherwise useful will be considered as serviceable. The serviceable/salvageable items (for which 'owner losing structure totally/partially' has no interest) will belong to the contractor, for which the contractor shall give credit to BMRCL (Summary Sheet)

3.3 **Unserviceable materials.** Any material declared by the Employer's representative are not serviceable shall be considered as unserviceable. A register shall be opened at the work site to show day-to-day account of the turnout of salvaged materials. The register shall also indicate whether dismantled materials are properly stacked or wasted. The contractor shall be responsible for the safe custody of all serviceable materials.

3.4 **Hazards in demolition.** Demolition of any structure is, inherently, more hazardous than the construction or erection of the same. From the point of view of safety, the conditions usually encountered while dismantling a structure, whatever its magnitude, do not lend themselves to the degree of control possible in the construction operations, where more stable conditions are generally obtainable. It is all the more imperative; therefore, that adequate attention is paid to planning and execution of demolition work, in its various stages, so as to minimize the risk of accidents and injuries to the personnel engaged in demolition operations. It has therefore become necessary to lay down certain safety procedures, which along with a planned programme could ensure adequate safety, particularly with the involvement of management, supervisors and workers. The demolition work shall be preceded in such a way that:

a) it causes least damage and nuisance to other parts of the same building, adjoining building and the residents of the building and the members of the public ; and

b) it satisfies all safety requirements to avoid mishaps and accidents.

3.5 **Planning.** Before beginning the actual work of demolition, the contractor shall carefully study the structure, which is to be pulled down and also of its surroundings. This shall include the following:

a) The manner in which the various parts of buildings are supported and how far the stage by stage demolition would affect the safety of the rest of the same structure and the adjoining structure;
b) A definite plan and procedure of demolition work shall be prepared, taking into account the loads on various structural parts and their supports;
c) Before commencement of each stage of demolition, the supervisor shall brief the workmen in detail regarding the safety aspects to be kept in view;
d) Ensure that the demolition conditions do not, at any stage, enhance the nuisance value of demolition work on the use of adjacent buildings;
e) No structure or part of the structure or any floor or temporary support or scaffold, side wall or any device for equipment shall be loaded in excess of the safe load bearing capacity, in its then existing condition; and
f) Stairs and stair railings, passage ways and ladders shall be left in place as long as possible. These should be maintained in a safe condition.

3.6 **Precautions and protective measures before starting demolition work**
The following precautions and protective measures shall be taken before commencement of demolition work:

(a) On every demolition job, danger signals shall be conspicuously posted all around the structure and all doors, openings giving access to structures shall be kept barricaded or manned except during the actual passage of workmen or equipment. However, provision shall be made for at least two independent exits for escape of workmen during any emergency.
(b) Walkways and passageways shall be provided for the use of the workmen who shall be instructed to use them and all such walkways and passageways shall be kept adequately lighted, free from all debris and other materials.
(c) Where in any work of demolition it is imperative, because of existing danger, to ensure that no unauthorized person shall enter the site of demolition outside working hours, watchmen shall be employed all 24 hrs.in a day. In addition to watching the site he shall also be responsible for maintaining all signs, notices, lights, barricades, etc. During nights, red lights shall be placed on or about the barricades.
(d) The power on all electrical service lines shall be shutoff and all such lines cut or disconnected at or outside the property line. The only exception would be any power lines required for the demolition work itself. Prior to cutting of/disconnecting such lines, the necessary approval of the Authority shall be obtained and beneficiaries intimated in advance.
(e) The structure to be demolished/dismantled must have barricades, which shall be of plain MS Sheet of 16 gauge fixed with steel frame, painting, with synthetic enamel paint of approved color, quality and brand, writing letter and logo of metro including maintenance of the same, duly cleaning the same frequently, arrangement for blinker light as per requirement. During demolition barricades has to be kept continuously. Removal of barricade will not be permitted till completion of demolition, dismantling of properties

(f) All mains and meters of the building shall be removed or protected from damage.
(g) All gas, water, steam and other service lines shall be shutoff and capped or otherwise controlled at or outside the property line.
(h) If a structure to be demolished has been partially wrecked by fire, explosion or other catastrophe, the walls and damaged roofs shall be shored and braced suitably.

(j) Construction sheds and toolboxes should be so located as to protect workers from injuries of falling objects, wall, etc.
(k) A warning device should be installed in the area to be used to warn the workers, in case of danger.
(l) Screens shall be placed, where necessary, to prevent flying pieces from injuring the fellow workmen.
(m) No demolition work shall be carried out during storm or heavy rain.
(n) No demolition work shall be carried out at night especially when the building or structure to be demolished is in an inhabited area.
(o) All necessary safety appliances shall be issued to the workmen and their use explained. It shall be ensured that the workers are using all the safety appliances while at work. The safety appliances should be as follows:

(1) Safety helmets as per IS 2925: 1984;
(2) Goggles made of celluloid lens to be worn at the time of demolition of floors, walls, tearing of plaster, etc., especially when equipment like jack hammers are used for demolition work, to protect the eyes from flying pieces, dust, dirt, etc. that may be blown up by wind.
(3) Leather or rubber gloves to be worn during demolition of RCC work or removing steel work, where the hands of workers are likely to be injured.
(4) Safety belts while working at higher level to prevent falling from the structure.

(p) First-aid equipment shall be available at all demolition works of any magnitude. Also, by prior arrangement, a qualified doctor be available at call.
(q) When there is a possibility of fire breaking out, appropriate portable first-aid fire appliances (see IS 2190: 1992) shall be kept at hand.
(r) The removal of a member may weaken the side wall of an adjoining structure and to prevent possible damage, these walls shall be supported until such time as permanent protection is provided. In case of any danger is anticipated to the adjoining structure, the same shall be got vacated to avoid any danger to human life.
(s) Ladders, when used, shall conform to IS 3696 (Part 2): 1991. Ladders or their side rails shall extend not less than 1.0 m above the floor or platform to which the ladder gives access. All ladders shall be secured against slipping out at the bottom and against movement in any direction at the top.
(t) All exterior wall openings which extend down to the floor level shall be barricaded to a height not less than 1 m above the floor level. All floor openings and shafts not meant as material chutes shall be floored over and endorsed with ground rails and toe boards.
(u) All existing fixtures/services required during demolition operations shall be well protected with substantial covering to the satisfaction of the Authority.
(v) When demolition is to be done by mechanical means such as weight ball and power shovels, the following additional precautions are necessary:

i. The area shall be barricaded for a minimum distance of 1 ½ times the height of the wall.

ii. The structure to be demolished/dismantled must have barricades, which shall be of plain MS Sheet of 16 gauge fixed with steel frame, painting, with synthetic enamel paint of approved color, quality and brand, writing letter and logo of metro including maintenance of the same, duly cleaning the same frequently, arrangement for blinker light as per requirement. During demolition barricades has to be kept continuously. Removal of barricade will not be permitted till completion of demolition, dismantling of properties

iii. While the mechanical device is in operation no workmen shall be allowed to enter the building being demolished;

iv. The device shall be so located as to avoid falling debris; and

v. The device when being used shall not cause any damage to adjacent structure, power line, other services, etc.

3.7 Protection of the public

Protection of the public before and during demolition is important and the following points should be kept in mind:

a) Every sidewalk or road adjacent to the work shall be closed or protected. All main roads, which are open to the public shall be kept open to the public clear and unobstructed at all times.
b) Children and public shall be kept out of the building and the adjoining yards.
c) If the structure to be demolished is more than two-storied or 7.5 m high, measured from the sidewalk or street which cannot be closed or safely diverted, and the horizontal distance from the inside edge of the side walk to the structure is 4.5 m or less, a substantial side walk shed shall be constructed over the entire length of the sidewalk adjacent to the structure of sufficient width with a view to accommodating the pedestrian traffic without causing congestion. The sidewalk shall be lighted sufficiently to ensure safety at all times.
d) The structure to be demolished/dismantled must have barricades, which shall be of plain MS Sheet of 16 gauge fixed with steel frame, painting, with synthetic enamel paint of approved color, quality and brand, writing letter and logo of metro including maintenance of the same, duly cleaning the same frequently, arrangement for blinker light as per requirement. During demolition barricades has to be kept continuously. Removal of barricade will not be permitted till completion of demolition, dismantling of properties.
e) A toe board at least 1 m high above the roof of the side walk shed shall be provided on the outside edge and ends of the sidewalk shed. Such boards may be vertical or inclined outward at not more than 45 degrees.
f) Except where the roof of a side walk shed solidly abuts the structure, the face of the sidewalk shed towards the building shall be completely closed by providing sheeting / planking to prevent falling material penetrating into the shed.
g) The roof of the sidewalk shed shall be capable of sustaining a load of 730 kg/m2. Only in exceptional cases, say due to lack of other space, the storing of the material on a sidewalk shed may be permitted in which case the shed shall be designed for a load of 1460 kg/m2. Roof of sidewalk shed shall be designed taking into account the impact of the falling debris. By frequent removal of loads it shall be ensured that the maximum load, at any time, on the roof of the shed is not more than 600 kg/m2. The height of the sidewalk shed shall be such as to give minimum clearance of 2.4 m.
h) Sidewalk shed openings, for loading purposes, shall be kept closed at all times except during actual loading operations.
i) The deck flooring of the sidewalk shed shall consist of plank of not less than 50 mm thickness closely laid and deck made watertight.
j) All members of the shed shall be adequately braced and connected to resist displacement of members or distortion of framework.
k) When the horizontal distance from the inside edge of the sidewalk to the structure is more than 4.5 m and less than 7.5 m, a sidewalk shed or fence may be built or in their place a substantial railing shall be constructed on the inside of the sidewalk or

roadway along the entire length of demolition side of the property with movable bars as may be necessary for the proper prosecution of the work.

l) Where workers entrances to the building being demolished are not completely protected by sidewalk sheds, all such entrances shall be protected by canopies extending from the face of the building to a point not less than 2.5 m from it. In such a case, overhead projection shall be at least 0.6 m wider than the building entrance or opening and every canopy shall be as strong as the sidewalk shed.

3.8 Sequence of demolition operations: The sequence of demolition shall generally be as given below:

a) The demolition shall always proceed systematically storey by storey in descending order and the demolition of upper floors shall be completely over before any of the supporting members or other important portion on the lower floor is disturbed. No unnecessary work shall go on below when the demolition is in progress above. When some work is to be done at the lower level, adequate protection shall be provided for all the workmen so engaged.

b) The requirements of (a) shall not prohibit the demolition of structures by sections, if means are taken to prevent injuries to persons or damage to property.

c) Roofs (or floors), generally, be demolished first before demolishing the supporting walls structural elements.

d) All glazed sash, glazed doors and windows etc., shall be removed before the demolition of roofs and walls starts. All fragile and loose fixtures shall be removed. Lath and loose plaster be stripped off throughout the entire structure. This is advantageous because it reduces glass breakage and also eliminates a large amount of dust producing material before more substantial parts of the building are removed.

e) The structure to be demolished/dismantled must have barricades, which shall be of plain MS Sheet of 16 gauge fixed with steel frame, painting, with synthetic enamel paint of approved color, quality and brand, writing letter and logo of metro including maintenance of the same, duly cleaning the same frequently, arrangement for blinker light as per requirement. During demolition barricades has to be kept continuously. Removal of barricade will not be permitted till completion of demolition, dismantling of properties

3.9 Demolition of floors. For demolition of floors the following procedure may be followed:

a) A slit in width not exceeding 300 mm shall be cut at the first stage for the entire length of the slab along which it spans. The opening shall thereafter be increased to the desired width by suitable installments.

b) Planks of sufficient strength not less than 50 mm thick and 250 mm wide shall be provided at a spacing not greater than 0.4 m. These planks shall be so placed as to give workmen firm support to guard against any unexpected collapse.

c) Stringers of ample strength shall be installed to support the planks where necessary and the ends of stringers shall be supported by floor beams, girders and not by floor slab alone.

d) When floors are being removed, no workmen shall be allowed to work in the area, directly underneath and such area shall be barricaded to prevent access to it.

e) The demolition of the floor in question shall be started only after the surrounding area for a distance of 6 m have been entirely cleared of persons, and the debris and other unnecessary material removed.

f) Planks used for temporary protection shall be sound and at least 50 mm thick. They shall be laid close together with ends overlapping at least 100 mm over solid bearing to prevent tipping under load.

3.10 Demolition of walls

3.10.1 Procedure. The following procedure shall be followed when demolishing walls:

a) While walls or sections of masonry are being demolished it shall be ensured that they are not allowed to fall as a single mass on the floors of the building so as not to exceed the safe carrying capacity of the floor; wherever practicable, they may fall away from the floors on to catch platforms. Overloading of floors shall be prevented by removing the accumulating debris through chutes or by other means immediately. The floor shall be inspected by the Authority before undertaking demolition work and if the same is found incapable of carrying the load of debris, necessary precautions shall be taken to prevent any unexpected collapse of the floor.
b) Walls shall be removed part by part. Stages shall be provided for the men to work on, if the walls are very thin and dangerous to work by standing over them.
c) No section of the wall whose height is more than 15 times the thickness, shall be permitted to stand without lateral bracing unless such a wall is in good condition and was originally designed to stand without such lateral bracing or support.
d) Structural or load supporting members on any floor shall not be removed or cut until all the storey above that floor have been demolished and removed.
e) Before demolishing any interior or exterior wall within 3 m of the opening in the floor immediately below, such opening shall be substantially planked over, unless access is denied to workmen to that portion of the floor immediately below the opening, in the floor of the storey being demolished, where any debris passing through the opening may fall.
f) In framed structures, if the structure is properly designed to the standards and to the specifications in such cases the frame may be left in position during demolition of masonry work. Where this is done all beams, girders, etc., shall be cleared of all loose materials as the demolition of masonry work progresses downward provided it is still strong enough to stand as an independent structure.

g) The structure to be demolished/dismantled must have barricades, which shall be of plain MS Sheet of 16 gauge fixed with steel frame, painting, with synthetic enamel paint of approved color, quality and brand, writing letter and logo of metro including maintenance of the same, duly cleaning the same frequently, arrangement for blinker light as per requirement. During demolition barricades has to be kept continuously. Removal of barricade will not be permitted till completion of demolition, dismantling of properties.
h) Walkways shall be provided to enable workmen to reach or leave their work on any scaffold or wall. Such walkways shall neither be less than 3 planks wide, nor less than 0.8 m in width.
i) After completion of each day's work, all walls shall be left stable to avoid any danger of getting overturned.
j) Foundation walls which serve as retaining walls to support the earth or adjoining structure, shall not be demolished until such an adjoining structure has been under pinned or braced and the earth removed by sheet piling or sheathing.

3.10.2 Catch platforms

Catch platforms shall be provided in case of demolition of exterior walls in multi-storey buildings. The following details may be considered:

a) Catch platforms shall generally be provided for multi-storied buildings more than 20 m high to prevent injuries to the worker and to the public when exterior walls are being demolished.
b) Such platforms shall be constructed and maintained not more than three storey below the storey from which the exterior wall is being demolished. When demolition has

progressed to within three storey of ground level, catch platforms will not be considered necessary.

c) Catch platforms shall not be less than 1.5 m in width measured in a horizontal direction from the face of the structure and shall consist of outriggers supported not more than 3 m apart. Planks shall be laid tight together, without openings between them and the walls. Catch platforms shall be provided with a continuous solid parapet along its outer edge of at least 1 m height. The parapet may be constructed with the same material as the platform.

d) Catch platform shall be capable of sustaining a live load of not less than 610 kg/m2.

e) Catch platforms shall neither be used for storing of materials nor dumping of materials.

3.11 Demolition of different types of structures and elements

3.11.1 General. Structures may be dealt with as masonry, concrete, steel and timber. The structures or their elements shall be dealt with as below, in addition to other requirements as applicable.

3.11.2 Masonry structures. Jack Arches – Where tie rods are present between main supporting beams, these should not be cut until after the arch or series of arches in the floor have been removed. Particular care should be exercised and full examination of structure be made before the demolition is commenced. The floor should be demolished in strips parallel to the span of arch rings (at right angles to the main floor beam).

Brick Arches

1) As much dead load as possible may be removed provided it does not interfere with stability of main arch rings; it should be noted that the load carrying capacity of many old arches relies on the filling between the spandrels. On no account should the restraining influence of the abutments be removed before the dead load of the spandrel fill and the arch rings are removed. The normal sequence of demolition includes the following:

- remove the spandrel filling down to the springing line,
- remove the arch rings,
- Remove the abutments.

2) A single span arch can be demolished, by hand, by cutting narrow segments progressively from each springing parallel to the span of the arch, until the width of the arch has been reduced to a minimum which can then collapse. Where it is impossible to allow debris to fall to the ground below, centering designed to carry the load should be erected and the arch demolished progressively. The design of the centering should make appropriate allowance for impact.

3) Where deliberate collapse is feasible the crown may be broken working progressively from the edges to the center.

4) In multi-span arches, before individual spans are removed, lateral restraint should be provided at the springing level. Demolition may be proceeded as for a single span care being taken to demolish the spandrels down to the springing line as the work proceeds.

3.11.3 Reinforced concrete

a) Before commencing demolition, the condition and position of reinforcement and possibility of lack of its continuity should be ascertained. Demolition should be commenced by removing partitions, non-load bearing cladding, etc. and similar non-structural elements.

b) Where hand demolition methods are used, the following procedures should be used:

1) **Beams –** For beams supporting rope should be attached to the beam. Then the concrete should be removed from both ends by pneumatic drill or cutting tools and the reinforcement exposed. The reinforcement should then be cut in which a way as to allow the beam to be lowered under control to the floor.
2) **Columns –** For columns reinforcement should be exposed at the base after restraining wire guy ropes have been placed around the member at the top. The reinforcement should then be cut in such a way as to allow the column be pulled down to the floor under control.
3) **Walls –** Reinforced concrete walls should be cut into strips and demolished as for columns.
4) **Suspended floors and roofs –** where ribbed construction is used, the principle of design and method of construction should be ascertained before demolition. Care should be taken not to cut the ribs inadvertently.

3.11.4 Pre-cast reinforced concrete

a) Pre-cast reinforced concrete units in a structure are normally held in position by the strength of the joints in-situ or on supporting walls, etc. As such before starting on demolition the joint structures or the supporting mechanisms shall be studied and understood.
b) In devising the following demolition sequences, due precaution shall be taken to avoid toppling over of the prefabricated units or any other part of the structure and wherever necessary temporary supports shall be provided.

3.11.5 Pre-stressed concrete. Before commencing of the demolition work involving such structures advice of an expert engineer should be obtained.

3.11.6 Steel

a) No beams shall be cut until precautions have been taking to prevent it from swinging freely and possibly striking any worker or equipment or any part of the structure being demolished.
b) All structural steel members shall be lowered from the building and shall not be allowed to drop.
c) Tag lines shall be used on all materials being lowered or hoisted up and a standard signal system shall be used and workmen instructed on the signals. No person shall be permitted to ride the load line.
d) When a derrick or hoisting equipment is used care shall be taken to see that the floor on which it is supported shall be strong enough for the loading. If necessary heavy planking shall be used to distribute the load to floor beams and girders. Overloading / overturning of the equipment shall be avoided.

3.11.7 Other elements

a) Roof trusses – Roof trusses shall be removed to wall plate level by hand methods. Sufficient purlins and bracing should be retained to ensure stability of the remaining roof trusses while each individual truss is removed. Temporary bracing should be added, where necessary, to minimize instability. The end frame opposite to the end where dismantling is commenced, or a convenient intermediate frame should be independently and securely guyed in both directions before work starts. On no account should the bottom tie of a truss be cut until the principal rafters are prevented from making outward movement.

b) Cantilevers – A cantilever type of construction depends on the balancing superimposed structure for its stability. Canopies, cornices, staircases, balconies should be demolished or supported before the balancing load is removed.
c) Heavy floor beams – Heavy bulks of timber should be supported before cutting at the extremities and should then be lowered to a safe working place.

3.12 Removal of materials

3.12.1 General. Removal of dismantled materials should be done carefully; they may be thrown / lowered to the ground. The materials shall preferably be dumped inside the building. Normally such materials shall be lowered to the ground or to the top of the sidewalk shed where provided by means of ropes or suitable tackles.

3.12.2 Through chutes

a. Wooden or metal shall be provided for removal of materials. The chutes shall preferably be provided at the center of the building for efficient disposal of debris.

b. Chutes if provided at an angle of more than 45 degree from the horizontal shall be entirely enclosed on all sides, except for opening at or about the floor level for receiving materials.
c. Opening for chutes shall not exceed 1.20 m in height measured along the wall of the chute and in all storeys below the top floor such opening shall be kept closed when not in use.
d. To prevent the descending material attaining a dangerous speed, the chute shall not extend in an unbroken line for more than two storeys. A gate or step shall be provided with suitable means of closing at the bottom of each chute to stop the flow of materials.
e. Chutes at an angle less than 45 degree to the horizontal may be left open on the upper side provided that at the point where such chute discharges into the chute steeper than 45 degree to the horizontal, the top of the steeper chute shall be boarded over to prevent the escape of materials.
f. Any opening into which workmen dump debris at the top of the chute shall be guarded by a substantial guard rail extending at least 1 m above the level of the floor or other surface on which men stand to dump the materials into the chute.
g. A toe board or bumper not less than 50 mm thick and 150 mm high shall be provided at each chute opening, if the required material is dumped from the wheel barrows. Any space between the chute and the edge of the opening in the floor through which it passes shall be solidly planked over.

3.12.3 Through openings

a) Debris may also be dropped through holes in the floor without the use of chutes. In such a case the total area of the hole cut in the intermediate floor, one which lies between floor that is being demolished and the storage floor shall not exceed 25 per cent of such floor area. It shall be ensured that the storage floor is of adequate strength to withstand the impact of the falling material.
b) Openings in all floors below the floor from which materials are being removed, shall be protected by standard railings and toe boards (see IS 4912: 1978) or preferably planked over if the holes are not being used for dumping materials.
c) All intermediate floor openings for passage of materials shall be completely closed with barricades or guard rails not less than 1 m high and at a distance of not less than 1 m from the edge of the general opening. No barricades or guard rails shall be removed until the storey immediately above has been demolished down to the floor line and all debris cleared from the floor.

d) When cutting a hole in an intermediate floor, between the storage floor and the floor which is being demolished, makes the intermediate floor or any portion of it unsafe, then such intermediate floor shall be properly shored. It shall also be ensured that the supporting walls are not kept without adequate lateral restraints.

3.13 References

Other Indian Standards on the subject of safety of workers are as follows:

IS No.	Title
3696 (Part 1): 1987	Safety code of scaffolds and ladders Part 1 Scaffolds
4014 (Part 2): 1967	Code of practice for steel tubular scaffolding ; Part 2 Safety regulations for scaffolding
3764: 1992	Code of safety for excavation work (first revision)
7969: 1975	Safety code for handling and storage of building materials.
13415: 1992	Code of safety for protective barriers in and around buildings.
13416 (Part-1): 1992	Recommendations for preventive measures against hazards at workplaces; Part 1 Falling material hazards prevention.
13416 (Part 2): 1982	Recommendations for preventive measures against hazards at work places; Part 2 Fall prevention.
13430: 1992	Code of practice for safety during additional construction and alteration to existing buildings.

PART – B: SPECIFICATIONS FOR REHABILITATION AND RESTORATION WORKS

Restoration works for safety reasons.

1. The BMRCL may appoint consultant/Engineer to conduct Building Condition Study (BCS) and suggest sequence and methodology of demolition including remedial measures required to be undertaken prior to demolition and/or during demolition and/or post demolition.

It is anticipated that, certain structures will be required to be constructed for safety and functional reasons as a result of Building Condition Study. The structures anticipated are front walls, doors, windows, beams and columns etc.

The BMRCL has identified few items which are to be executed in this contract. These items shall be as per KPWD Common Schedule of Rates for the year 2023-24. Measurements will be recorded for only those buildings completed under the supervision of consultant/Engineer appointed for Building Condition Study.

2. All Standards, Technical Specifications and Codes of Practice referred to shall be latest editions including all applicable official amendments and revisions. The contractor shall make available at site all relevant KPWD Specifications and Indian Standard Codes of practice as applicable.

In case of discrepancy among Standard codes of practice, Technical Specifications and provisions in Sub clauses in this 'Employer Requirement, Scope of Work & Technical Specifications', the order of precedence will be as below:

i) Provision in this 'Employer Requirement, Scope of Work & Technical Specifications'
ii) Technical Specifications
iii) KPWD Specifications
v) MORTH Specifications
vii) Standard Codes of Practice

3. The utilities which are not to be diverted but require supporting, proper supporting be done so that they are not damaged along their branches. All Precautions shall be taken while handling the utilities. The Contractor shall be liable for any damages in this behalf.

4. Before carrying out the work at site, necessary permissions as required from various local agencies / authorities shall be required to be obtained by the contractor. The Employer shall assist only by way of issue of necessary support letters.

5. The work has to be executed carefully without damaging the existing surface/over ground/underground utilities. The contractor shall execute electrical and plumbing works through Licensed Electrical/Plumbing contractors.

6. The Contractor shall strictly adhere to the working drawings or detail drawings and written instructions of the Engineer.

TECHNICAL SPECIFICATIONS

1. MATERIALS AND WORKMANSHIP: The whole of the materials used on the work shall be new and of the best quality of their respective kind. Other articles, tools, plants shall be the best of the several kinds available in the market and shall be used in the best, most workman like, and most substantial manner to the complete satisfaction of the Engineer.

The workmanship is to be the best available and of a high standard. The Contractor shall take all steps immediately to make up deficiency if any noticed by the Engineer. Use must be made of a special tradesman in all aspects of the work and allowances must be made in the rates for the same.

Contractor shall maintain uniform quality and consistency in workmanship throughout the execution of the work.

Materials shall be of the best approved quality obtainable and they shall comply with the respective Indian Standard Specification.
In case of non-availability of materials in metric sizes, the nearest size in FPS units shall be provided with the prior approval of the Engineer for which neither extra will be paid nor shall any rebate be recovered.
It shall be obligatory for the Contractor to furnish Certificate, if demanded by the Engineer, from manufacturer or the material supplier that the work has been carried out by using their material and as per their recommendation.
Unless otherwise shown on the drawings or mentioned in the schedule of quantities or special specifications the quality of materials, workmanship, dimensions, etc., shall be as specified here in under.
All equipment and facilities for carrying out field tests on materials shall be provided without any extra cost.

2. TESTING OF WORKS AND MATERIALS: If required by the Engineer/Employer the Contractor shall arrange to test materials, concrete cubes etc. and portions of the work in order to prove their soundness and sufficiency at his own cost. If, after any such test and in the opinion of the Engineer the materials or workmanship is found defective or unsound, the Contractor shall pull down and re-execute the same at his own cost. All materials deemed defective in the opinion of the Engineer shall be removed from the site immediately.

3. SAMPLES OF MATERIAL: The Contractor shall deposit at the office of the Engineer for their approval samples of the materials proposed to be used in the works, at least 15 days before placing the order supply, and the bulk shall be equal in every respect to the approved samples deposited.

4. STORAGE: Safe, dry and proper storage shall be provided for all materials, particularly for materials like cement. Adequate storage should be made at least sufficient for 1 month during active construction.
The Contractor shall procure necessary guarantees from the manufactures and furnish the same to the Engineer.

5. SPECIFICATIONS FOR CONCRETE WORKS

Concrete General
Wherever grade of concrete like M15, M20 etc. are mentioned, the concrete mix shall be designed by the contractor and approval of Engineer for the same shall be obtained. Weigh batching shall be adopted for producing the concrete. Obtaining of concrete from concrete batch mix plant will be permitted. Contractor shall obtain approval of Engineer for each such batching plant. Proportion of ordinary cement concrete will be expressed as I:3:6, 1:2:4, 1:1.5:3 etc. in the specification. The first figure will be the quantity of cement by volume, the second figure will be dry coarse sand (fine aggregate) by volume and the third figure will be the quantity of course aggregate by volume. Cement shall be measured by weight. The weight is to be derived on the basis that 1 cubic meter of cement will weigh 1440 kgs., or one full bag of 50 kgs. will be assumed to be 35 ltrs. when the sand is wet or moist suitable correction for bulking is to be given while proportioning. Engineer may allow measuring cement by volume.
Unless otherwise specified the rates for all RCC will be exclusive or reinforcements. Reinforcements will be paid for separate1y.
Unless, otherwise stated, for all RCC work the size of course aggregate will be 20 mm and down size.

6. CEMENT CONCRETE PLAIN AND REINFORCED:

A) Cement: Cement shall comply in every respect with the requirements of the latest publication of IS: 8112 and unless otherwise specified, ordinary Portland cement shall be used. No other make of cement but that approved by the Engineer will be allowed on works and the source of supply not be changed without approval of the Engineer in writing. Test certificates to show that the cement used fully complies with the relevant IS specifications shall be submitted to the Engineer and not withstanding this, the Engineer may at their discretion order that the cement brought to site and which they may consider damaged or of doubtful quality for any reasons whatsoever shall be re-tested in an approved testing laboratory and fresh certificate of its soundness shall be produced. Cement ordered for retesting shall not be used for any work pending results of retest.

Cement shall be stored and neatly packed in piles not exceeding 10 bags high in weather proof sheds with raised wooden plank flooring to prevent deterioration by dampness or intrusion of foreign matter. It shall be stored in such a way as to allow the removal and use of cement in chronological order of receipt, i.e. the first received being first used. Cement deteriorated and/or clotted shall not be used on work but shall be removed at once from the site.

B) Fine Aggregates: Sand shall conform to IS: 383. It shall pass through 1.5 sieve 4.75 mm (3/16 B.S) test sieve leaving a residue not more than 5%. It shall be from a natural source or crushed stone screedings. It shall be washed if directed to reduce the percentage of deleterious substances to acceptable limits. Sand shall not contain any trace of salt and sand containing any trace of salt shall be rejected.

The fine aggregate for concrete shall be graded within limits as specified in IS: 383 and the fineness modules shall range between 2.60 to 3.20. The fine aggregate shall stacked carefully on a clean hard dry surface so that it will not get mixed up with deleterious foreign materials. If such a surface is not available a platform of planks or corrugated sheets or brick floor or concrete floor shall be prepared. Sand shall be added in the desired proportion as required for the strength specified, with suitable correction for bulking.

C) Coarse Aggregates: Coarse aggregate shall conform to IS: 383. It shall consist of crushed or broken stone, 95% of which shall, be retained on 4.75 mm is test sieve. It shall be obtained from crushed granite, trap, basalt or similar approved stones from approved quarry. Coarse aggregate shall be chemically inert when mixed with cement & shall be angular in shape and free from soft friable thin porous laminated or flaky pieces. It shall be free from dust and other foreign matter. Gravel/shingle of desired grading may be permitted as a substitute in part or full in plain cement concrete if the Engineer are otherwise satisfied about the quality of aggregate.

D) Water: Water used for mixing concrete shall be fresh, clean, free from oil, salts, acids, alkali and shall be in accordance with the clause 4.3 of IS 456 - 1978. The contractor shall produce test results for approval by the Engineer for the mixing of water proposed to be used on the job, before casting any concrete.

E) Mixing Of Concrete: MACHINE MIXED: Aggregates shall be accurately measured out in boxes and mixed dry along with required cement, water shall then be added in measured quantity and mixing shall be continued until there is uniform distribution of the materials and the mass is uniform in color and consistency but in no case shall the mixing be done for less than 2 minutes. Only hopper loading mixer shall be used.

7. CONSISTENCY:

Only sufficient water giving due allowance for the moisture content of aggregate shall be added to the cement and aggregates during mixing to produce a mixture of sufficient workability to enable it to be well consolidated, to be worked into the corners of the shuttering and around the reinforcements where there is reinforcement to give the specified finish and to have specified strength.

Normally for every 50 kg of cement total water including moisture content of aggregate should not be more than 34 liters for 1:3:6 mix 32 liters for 1:2:4 mix. 30 liters for 1:1½:3 and 27 liters for 1:1:2.

If difficulty be experienced in placing the concrete of specified mix and approved consistency between and below reinforcement bars, in the bottom of beams and similar situations, the concrete shall have improved workability by increasing the proportion of water and corresponding additional quantity of cement and using aggregates of smaller size than specified as directed by Engineer for which no extra be paid.

The consistency shall be determined by making trial mixtures with dried aggregates, or when so instructed by test of laboratory made test cubes under the direction of Engineer. Consistency may be measured by slump test using a standard cone or the Engineer may direct the use of any other means of testing the consistency.

If the apparatus used for the slump test is a standard cone when filled. It shall be raised vertically clear of the concrete and the measurement of the slump shall be 300 mm minus the height of the slumped cone of concrete. Care shall be taken to prevent vibration of the samples being tested. The following slumps shall be adopted for different kind of works:

WITH/WITHOUT VIBRATOR

1. Mass concrete in RCC foundation, footings & retaining walls. 10 mm to 25 mm 80 mm
2. Beams, slabs and columns 25 mm to 40 mm 100 mm to 125 mm simply reinforced
3. Thin RCC section or 40 mm to 50 mm 125 mm to 150 mm section with congested steel

8. PLACING AND COMPACTING: Method of placing shall be such as to preclude segregation and as far as practicable the placing shall be continuous.

Special care shall be taken in accordance with IS 456 while laying concrete under extreme weather. Concrete, during the operation of placing shall be thoroughly worked around the reinforcements, embedded fixtures, spaded against corners of the formwork by punning, rodding, or by any other approved means and thoroughly compacted by mechanical vibrators. The number and type of vibrator to be used shall be subject to the approval of the Engineer, and in general immersion type vibrators shall be used.

9. TRANSPORTING: Concrete shall be conveyed from the place of mixing to the place of final deposit as rapidly as practicable by methods which will prevent segregation or loss of any of the ingredients. If segregation does occur during transport, the concrete shall be remixed before being placed. Normally not more than 30 minutes shall lapse between mixing and consolidation in position.

10. CURING: All cement concrete after laying shall be protected from damages, till it sets and shall be cured thereafter for not less than 10 days. The work shall be protected from drying wind and direct sun rays. Water used for curing shall be free from sediments of any kind and generally fit for drinking.

11. CONSTRUCTION JOINTS: Construction joints in exposed concrete work shall be made only where shown on the drawings or directed by the Engineer and shall be in accordance with the details shown or approved by the Engineer. The procedure given in clause 20.1.4.2 of IS; 456-1978 shall be followed for general guidance. All laitance shall be removed from the concrete before it is allowed to fully harden. The removal shall be

effected by scrubbing the concrete surface with wire and with bristle brushes and washing down to expose clearly the aggregate. However, care shall be taken to avoid dislodgement of particles of aggregate.

If concrete has been allowed to harden excessively the surface shall be chipped over its whole surface to a depth of at least 10 mm and thereafter thoroughly washed. Before fresh concrete is added on the construction joints, the surface of the old concrete shall be thoroughly wetted and covered with a thin layer of cement mortar 1:2.

12. VERTICAL JOINTS: At the end of any day's work or run of concrete, the concrete shall be finished of against temporary shutter stop which shall be vertical and securely fixed. This stop shall be removed as soon as the weather permits.

13. STRENGTH OF ORDINARY CONCRETE: The contractors have to see that proper materials are used and the proportion and the correct water cement ratio just sufficient for the workability are maintained to see that the following minimum strength of concrete are obtained.

To verify this, test cubes from the concrete used should be made and tested. The frequency of testing and the acceptability criteria will be according to IS 456 and IS 516.

14 CONCRETE FORMWORK AND CENTERING:

14.1 General:

Form work shall be rigidly constructed of 12 mm thick new plastic coated plywood sheets for column, pedestals, beams, walls and fascia and steel or plywood for roof slab, canopies, floors, cover slabs, retaining walls, etc. Shuttering faces in contact with the concrete shall be free from adhering grout, projecting nails, splits or other defects that are likely to mar the concrete surface. The shuttering shall be erected on battens, beams and steel props properly cross-braced so as to make the form work rigid.

Form work shall be erected true to line and levels and to the required shapes as called for and shall be able to carry without deformation the full weight of wet concrete and live loads. It shall also withstand the effect of vibration without deflection, bulging, distortion or loosening of its component parts. Details of centering and form work shall be subject to the approval of the Engineer. The completed form work shall also be subject to the approval by the Engineer before placement of reinforcement. Contractor shall be responsible for sufficiency and adequacy of all form work centering and moulds.

All joints shall be sufficiently tight to prevent leakage of grout. Chamfer fillets shall be provided at all corners wherever called for on the drawings or as directed by the Engineer. Openings may .be given with the approval of Engineer at convenient places for washing down all the rubbish. These are to be closed before concreting.

14.2 Surface Treatment of Shuttering: The surfaces of shuttering coming into contact with concrete shall be plastic coated of approved make. The shuttering shall be thoroughly cleaned and coated before each use.

14.3 Removal of Form Work: All formwork shall be removed without shock or vibration and shall be eased off carefully in order to allow the structure to take up its load gradually. Forms shall not be disturbed until, concrete had adequately hardened to take up its own weight and superimposed load coming on it and in no circumstance shall forms be struck until the concrete reaches strength of at least twice the stress to which the concrete may subjected at the time of striking. The side forms shall be so fixed that while removing them the supporting forms & posts are not disturbed.

14.4 Surface Finish of Concrete:

Only cement mortar rendering of maximum 6 mm thickness may be permitted as finishing to concrete surface. Imperfect surfaces shall not be patched but shall be repaired by removal and replacement of the member as directed by the Engineer.

If centering and shuttering is to be paid for separately, measurement will be taken according to IS 1200. The unit area for payment will be one square meter.

All aggregate are to conform strictly to IS specification IS 383. The aggregates will be tested as frequently as directed by the Engineer to see that their specification is the same as adopted in the mix design. They must be stored on clean platform made for the purpose.

Concrete shall be weighed batched. The dials of weigh batching unit to be used shall be checked with standard weights periodically. The conversions of weights to volume will be allowed by the Engineer under special circumstances. Despite the design for several mixes the following quantities of cement are the minimum to be used for the different grades of the concrete:-

Cement concrete	(1:4:8)	- 3.40 Bags per cum.
Cement concrete	(1:3:6)	- 4.40 Bags per cum.
Cement concrete	(1:2:4)	- 6.40 Bags per cum.
Cement concrete	(1:1.5:3)	- 8.00 Bags per cum.
RCC	(1:1.5:3)	- 8.00 Bags per cum.
RCC	(1:1:2)	-10.80 Bags per cum

14.5. DEFECTS IN CONCRETE:

Immediately on removal of form work, the surface of the concrete shall be examined by the contractor and any bone combs or other imperfections shall be brought to the notice of the Engineer. The acceptability or otherwise of such defective concrete shall be at the sole discretion of the Engineer who may direct the contractor to repair the defective work or ask for demolition and replacement of such defective work at the risk and cost of the contractor.

CONCRETE WORK
LIST OF BUREAU OF INDIAN STANDARDS

Sl No	IS No.	Subject
1	383-1970	Coarse and fine aggregate from Natural source for concrete (2nd revision) Reaffirmed 1990.
2	456-2000	Code of practice for plain and reinforced concrete (3rd revision) (Amendments 2) Reaffirmed 1991.
3	516-1959	Method of sampling and analysis of concrete. Reaffirmed 1991.
4	1200 (Part II) 1974	Method of measurement of building and civil engineering work Part 2 (concrete works). (3rd revision) (Amendments 2) Reaffirmed 1991.
5	1322-1993	Bitumen felt for water proofing and damp proofing (4th revision) (previously 13220-1982)
6	1786-1966	High strength deformed steel bars and wires for concrete reinforcement.
7	1791-1985	Batch type concrete mixers. (2nd revision) Reaffirmed 1990.
8	2386-1963	Method of test for aggregate for concrete work. a) Part 1 particle size and shape (Amendments 2) Reaffirmed 1990 b) Part 2 Estimation of deleterious materials and organic impurities (Amendments 1) Reaffirmed 1990. c) Part 3 Specific gravity, density, voids, absorption and bulking – Reaffirmed 1990.l d) Part 4 Mechanical properties (Amendments 3) Reaffirmed 1990.
9	2505-1980	General requirements for concrete vibrators immersion type. Reaffirmed 1993.
10	2505-1985	General requirements for screed board concrete vibrators. (1st revision) Reaffirmed 1990.
11	2645-1975	Integral cement water proofing components (1st revision) (Amendments 1) Reaffirmed 1992.
12	2686-1977	Cinder as fine aggregate for use in lime concrete (1st revision) (Amendments 1) Reaffirmed 1992.
13	3812-1981	Fly ash for use as Pozzolana and admixtures (1st revision) Reaffirmed 1992.
14	4656-1968	Form vibrators for concrete. Reaffirmed 1991.
15	8112	43 grade OPC
16	9103-1979	Admixture for concrete Reaffirmed 1990.

15. REINFORCEMENT: Steel reinforcement shall be either mild steel of tested quality high yield strength deformed bars of grade Fe-415 conforming to IS; 1786--1966, Fe-500 or as called for on the drawings Fabric reinforcement where called for shall be of hard drawn mild steel wire mesh conforming to IS: 1566-1967. Bars shall be free from deleterious materials, mill scale, loose rust, oil or paint. The contractor shall submit bar bending schedules for approval of the Engineer prior to commencement of fabrication. These shall indicate the accurate dimensions and bending of bars as called for on the structural drawings. Fabrication shall be accurately done to the dimensions, spacing and ensuring minimum cover as called for on structural drawings. All steel shall be rigidly held in place with 18 gauge annealed steel wire. Cement mortar (1;2) blocks of required shape, M.S. chairs and spacer bars shall be used in order to ensure accurate positioning of reinforcement.

All reinforcing steel within the limit of a day's polish shall be in place and firmly wired at least one day, prior to the date of pour to permit inspection. The contractor shall also ensure that all conduits and inserts are in position before placing concrete.

Hooks where indicated shall be either a complete semi-circular turn with a radius of not less than four and not more than six bar's diameters, plus an extension of at least four bar diameters at the free end, or a 90-degree bend having a radius of not less than 4 bar diameters plus an extension of 12 bar diameters. Unless otherwise shown or specified on the drawings, all lapped by 50 times the diameter of bar. Not more than 33%. (Thirty-Three Percent) of the bars shall have lapped joints at the same location. Only where specifically shown on the drawings, reinforcement shall have welded joints. Rate quoted for steel reinforcement work shall be deemed to include the cost of such welding wherever specified.

16. STACKING AND STORAGE: Steel for reinforcement shall be stored in such a way as to prevent distorting and corrosion. Bars of different classifications, sizes and lengths shall be stored separately to facilitate issue in such sizes and lengths to cause to minimum wastage in cutting from standard length.

17. EARTHWORK INCLUDING GENERAL FILLING, DEWATERING AND BACKFILLING:

These Specifications cover the general requirements for earthwork and related works in different types of soils and rocks. The scope of work includes, but is not limited to, clearing of the site; general filling of site; compaction of fill to required degree; excavation and backfilling works; works related to stabilizing of sides of excavations; and necessary dewatering all within the intent of these specifications. The scope shall include provision of access to different parts of the excavated areas, adequate protective safety measures, and proper lighting and necessary signage arrangements.

General filling using approved material shall be done in areas as indicated in drawings or as directed by the Engineer.

The tenderer should acquaint him with the site for assessing the efforts necessary for completing the work as per these specifications.